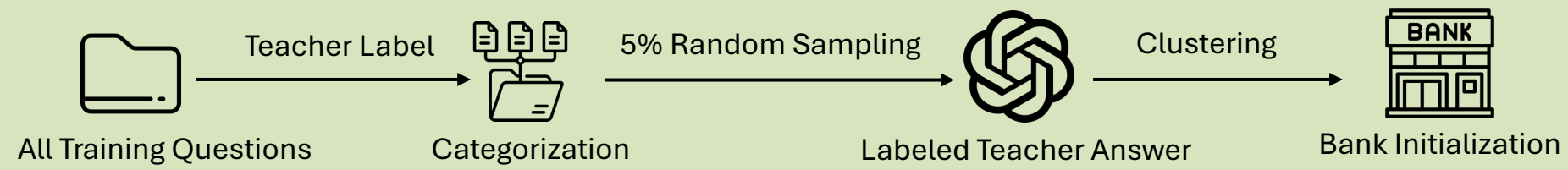
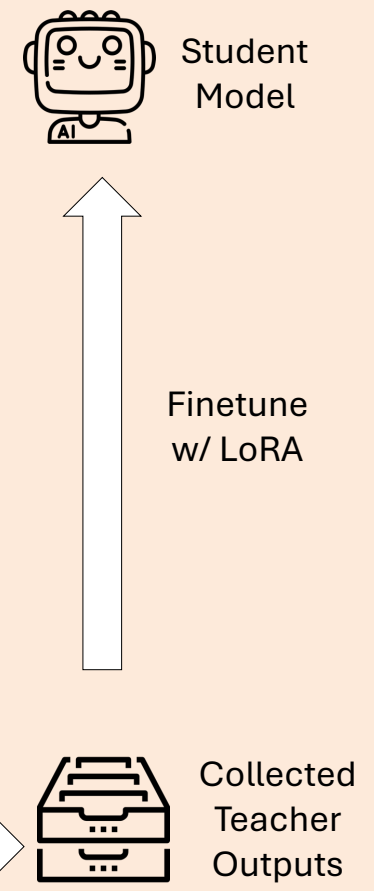


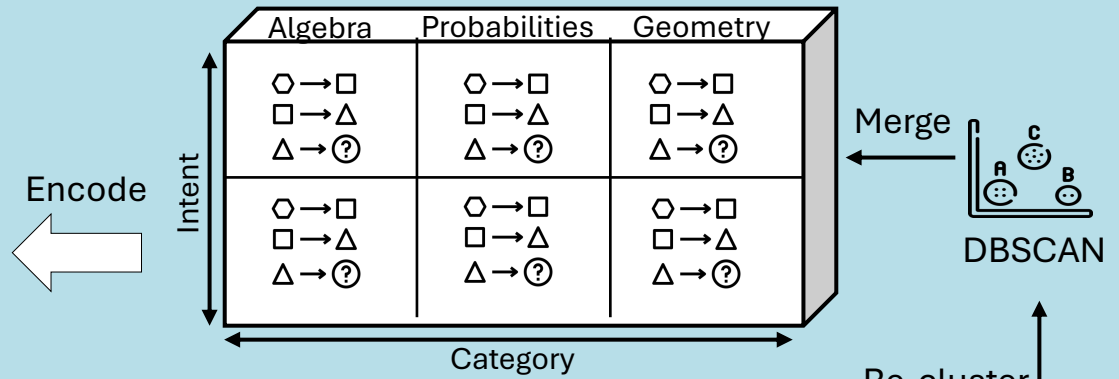
Stage 1: Initial Bank Construction



Stage 3: Student Model Training



Reasoning Path Bank



Encode

Q: Tim buys 3 dozen eggs. Eggs cost \$.50 each. How much did he pay for eggs?

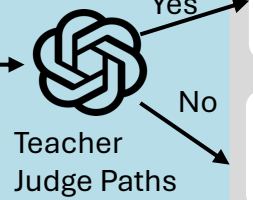
Question Embedding

Intent Embedding

Top-K Retrieval (k = 2)

Path 1: ComputeIndividualTotals -> CombineTotals

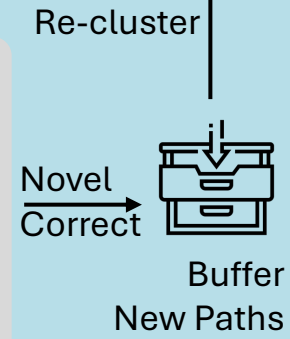
Path 2: CombineAmounts -> SubtractTotalCost



JSON Output

Rationale: ComputeIndividualTotals: $12 * 3 = 36$. CombineTotals: $36 * 0.5 = 18$. Answer: 18\$

LLM Generated Paths + Rationale + Answer



Stage 2: SFT Reasoning Extraction